

Reviewer Recommendation and Comments for Manuscript Number CSEM-D-25-02918**An agent-based study on the dynamic distribution and firms concentration in a closed economy**

Original Submission
 Thanh Binh Lai, Ph.D candidate **Reviewer 3**

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If this submission is transferred to another publication, do we have your consent to include your identifying information?

Yes

If this submission is transferred to another publication, do we have your consent to include your original review?

Yes

Reviewer Comments to Author

Thank you for the opportunity to read your manuscript. The topic of firm dynamics and market concentration is interesting, and I appreciate the effort you put into combining simulations, analytical derivations, and sectoral extensions. After careful consideration, however, I believe the manuscript requires substantial redevelopment before it could be suitable for publication, and I outline below several issues that you may find helpful in revising or redirecting your work.

1. Clarification of the modeling framework: The paper is presented as an agent-based model, yet firms evolve independently as multiplicative stochastic processes without interaction. In the standard literature, agent-based economic models rely on interaction among agents to generate aggregate outcomes (Tesfatsion & Judd 2006; Macal & North 2010). Clarifying this distinction and perhaps framing the model as a multi-agent stochastic growth process would help position the work more accurately.

2. Interpretation of "business cycles": The manuscript suggests that the simulated paths display business-cycle-like fluctuations. In macroeconomics, business cycles are typically defined as recurrent co-movements of aggregate variables driven by identifiable shocks and propagation mechanisms (Burns & Mitchell 1946; Lucas 1977; Kydland & Prescott 1982). Since the present model generates independent random-walk trajectories without co-movement or structural shocks, it may be clearer to avoid the business-cycle terminology or to reinterpret the results more cautiously.

3. Limited novelty relative to established stochastic growth models: Several key findings—such as the emergence of dominant firms under high drift or the reduction in concentration under parameter heterogeneity—follow directly from the mathematical structure of multiplicative random walks. These outcomes arise from exponential compounding rather than from additional economic mechanisms. Developing additional mechanisms, introducing interaction, or linking to empirical patterns would strengthen the economic contribution.

4. Minor comments:
 Acronym inconsistency (HHI vs. HMMI vs. HHIM vs. MHHI) should be corrected.
 In-text citations contain some inconsistencies.
 Figures would benefit from clearer axes, labels and captions.

Overall, I hope these comments assist you in refining your modeling approach and the presentation of your results. Thank you again for the opportunity to review this work.

Reviewer Confidential Comments to Editor:

1. Is the paper acceptable for publication?
 The paper is not acceptable for publication in its present form, nor would minor or major revisions be sufficient to bring it to the standard expected for Computational Economics.
 I therefore recommend Reject.

2. Rationale for Recommendation

The manuscript's central limitations are structural rather than incremental:

- The model consists of independent multiplicative random walks with no interaction among firms, yet it is framed as an agent-based model. Under standard definitions (Tesfatsion & Judd 2006; Macal & North 2010), agent-based computational models require decentralized interactions among agents. This fundamental mismatch limits the methodological contribution.
- Claims about generating "business cycles" are not supported. The model lacks structural shocks, co-movement, propagation mechanisms, or any macroeconomic structure that defines business-cycle dynamics (Burns & Mitchell 1946; Lucas 1977; Kydland & Prescott 1982). The simulated paths represent random-walk fluctuations rather than economic cycles.
- Several key results (e.g., dominance of firms with higher drift, reduced concentration with heterogeneous parameters) follow mechanically from the multiplicative growth assumption and therefore offer limited economic novelty.
- Additional methodological issues—including an unspecified price process, limited sensitivity analysis, and underdeveloped regression methodology—further constrain the publishability of the paper.

Given these structural issues, I do not believe the manuscript could reach the journal's standards through revision

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